CT45: exact finite-level cross-factor Hom

E109 / E113 over F101 | 2026-11-12 | [pcc:unverified]

Theorem in Frobenius companion bases

For m coprime to 101: C(t) = [[0,-101],[1,t]] mod m.

Every X satisfying X C(-7) = C(-11) X has exactly the form:

X(a,c) = [[a,-101c],[c,a-11c]], with 4a=4c=0 (mod m).

Thus 4X=0. Odd levels have no nonzero cross-factor maps.

For m=2^k, there are exactly 16 intertwiners at all k >= 2.

Independent check results

Modulus	All maps	Units	Main observation
2	4	3	three 2-torsion maps
4	16	12	six Weil anti-isometries
8	16	0	every entry even
16	16	0	every entry divisible by 4
32/256	16 each	0	formula-checked; no brute claim

Interpretation and scope

No level-four anti-isometry lifts to a full equivariant 8-torsion isomorphism.

Finite-level Galois maps need not be geometric elliptic homomorphisms.

No actual genus-two kernel, PCC signature or DLP speedup is established.

Controls: independent Python/JS exhaustive mod 2/4/8/16; positive trace -15.

Sources: Bruin-Doerksen (2011), crypto PR #3877, cryptanalysis PR #1502.

The 4-torsion lift barrier

Level 4	Level 8	Level 16
12 invertible	0 invertible	0 invertible
6 anti-isometries	all entries even	all entries / 4

Every Frobenius intertwiner is annihilated by 4.

Proof: expand X C(-7)=C(-11) X. The trace gap is four.

The resulting two parameters a,c satisfy 4a=4c=0 modulo m.

Therefore the only compatible 2-adic Tate-module map is zero.

Positive mutant: trace -15 admits 48 invertible maps mod 8.

Next: compare the actual native U graph, degree-7 A/B and CT44 dual using independent Cantor and Weil tests. Full solver costs still pending.

Reproduce: python3 -B .../ct45_cross_torsion.py

node .../ct45_cross_torsion.js

[pcc:unverified] | No paid compute.